

TESTs - Lesson 16 - Coupling and Oracles

What is the primary purpose of Oracles in blockchain?

- A To process on-chain transactions faster.
- B To allow smart contracts to access and use off-chain data.
- C To secure private keys on the blockchain.
- D To create new cryptocurrencies.

Why is a backend often required for tasks like token minting in a dApp, rather than performing it directly on the frontend?

- A To improve the user interface and experience.
- B To keep private keys secure and implement complex business logic.
- C To reduce gas fees for blockchain transactions.

What is the key difference between GET and POST requests in a REST API, especially regarding idempotency?

- A GET requests always require a body, while POST requests do not.
- B POST requests are idempotent, meaning multiple identical requests have the same effect; GET requests are not.
- C GET requests are idempotent, meaning multiple identical requests have the same effect; POST requests are not.

What is a DTO (Data Transfer Object) primarily used for in the context of REST APIs as explained in the video?

- A To create new database entries.
- B To manage server-side routing logic.
- C To define the structure of data sent between processes, especially for request bodies.
- D To encrypt data transmitted over the network.

What does CORS stand for, and why is it a common issue when coupling frontend and backend applications?

Starting...



- A Centralized Object Request System; it prevents backend servers from crashing.
- B Cross-Origin Resource Sharing; it is a security feature preventing unauthorized cross-domain requests.**
- C Client-Side Object Rendering Standard; it simplifies UI development.

What is a crucial reason to use a backend for token minting, rather than doing it directly from the frontend?



- A Frontends are not powerful enough to execute blockchain transactions.
- B To keep the minter's private keys secure and handle complex business logic (like verifying social media posts for airdrops).**
- C To improve the visual aesthetics and user experience of the dApp.

What are the main advantages of using a "loose coupling" architecture between frontend and backend services, as discussed in the video?



- A Increased performance for blockchain transactions and reduced gas fees.
- B Simplified development by using a single programming language across the entire application.
- C Flexibility in language choice, independent deployment, easier maintenance, and improved scalability.**

What is the overall architectural principle emphasized for building dApps involving frontend and backend.



- A Strong coupling, where frontend and backend components are tightly integrated into a single framework.
- B Loose coupling, where frontend and backend services communicate via standard protocols (like HTTP) allowing independent development and deployment.**
- C Centralized architecture, where all logic resides on a single server for maximum control.

How does Tellor address the "Oracle Problem" and ensure data integrity in its network, as described in the video?



- A By allowing only pre-approved, centralized entities to provide data.
- B By implementing a permissionless network where anyone can request, fund, and report data, relying on economic incentives and a dispute mechanism.
- C By directly integrating all off-chain data sources onto the blockchain.